

Data Science Project Report--Data-Driven Strategies for Reducing Telecom Customer Churn

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Content

1. Executive Summary

(1) Background

Telecommunications companies have a high customer churn rate and urgently need to identify the key factors affecting churn and propose retention strategies.

(2) Key findings

① The churn rate of long-term contract customers is significantly lower than that of short-term contract customers.

② The use of value-added services such as technical support and online security is highly correlated with customer retention.

③ Customers with high monthly fees are more likely to churn, especially when there is a lack of additional service support.

(3) Suggestion

① Promote long-term contracts and preferential packages.

② Strengthen the dissemination of technical support and value-added services.

③ Design personalized retention plans for high-monthly-fee customers.

2. Problem Definition

(1) Business question

How to predict whether customers will churn and identify the main influencing factors?

(2) Objectives and success criteria

Build a predictive model with an accuracy of $\geq 75\%$ and provide actionable business insights.

(3) Key issues

① Which variables best explain customer churn?

② Can the model identify high-risk customers in advance?

③ How can analysis results be transformed into retention strategies?

3. Data Strategy

(1) Data source

Telecom customer churn dataset provided by IBM (public data).

(2) Data quality

The original data contained 7043 records and 21 variables. Missing values appeared only in TotalCharges (11 records) and have been removed.

(3) Preprocessing steps

① Delete the useless field (customerID).

② Convert the target variable Churn to binary.

③ Perform One-Hot encoding on categorical variables.

④ Standardize numerical variables.

(4) Key findings of EDA

- ① The overall attrition rate was approximately 26.6%.
- ② Contract type, payment method, and use of technical support are highly correlated with churn.
- ③ Customers with higher monthly fees have a higher churn rate.

4. Analysis & Methods

(1) Model selection

Logistic regression and random forest were considered. Logistic regression was ultimately chosen due to its strong interpretability.

(2) Feature selection

Retain all service, contract, payment method, and fee-related variables.

(3) Evaluation metrics

accuracy, precision, recall, F1 score, ROC, AUC.

(4) Validation method

Training set/test set split (70/30).

(5) Adjusting parameters for optimization

Increase the number of iterations and use StandardScaler to standardize the data.

5. Insights & Recommendations

(1) Key results

- ① The model accuracy is approximately 80%, and the AUC is approximately 0.71.
- ② The identification rate of non-churned customers is high (recall rate 89%), while the identification rate of churned customers is moderate (recall rate 54%).

(2) Business value

The model can help companies identify some high-risk customers in advance and take targeted measures.

(3) Risks and limitations

The churn customer identification rate is limited, and some high-risk customers may be missed.

(4) suggestion

- ① Short-term: Promote long-term contracts and improve technical support services.
- ② Long-term: Optimize pricing strategies and improve model recognition capabilities by incorporating customer satisfaction surveys.

6. Evaluation & Future Improvement

(1) Limitations

The data comes from only a single telecommunications company, and external factors are not considered.

(2) Areas for improvement

Incorporate more customer satisfaction and usage behavior data.

(3) Ideal project vision

To achieve dynamic churn prediction by combining real-time data streams with more complex models (such as XGBoost).

7. Learning & Reflection

(1) Course takeaways

Mastered the complete data science process (EDA → Modeling → Evaluation → Insight).

(2) Self-learning achievements

I have learned pandas, seaborn, and scikit-learn in Python.

(3) Challenges and solutions

Issues such as missing values, missing libraries, and model convergence were encountered and resolved step by step.

(4) Future applications

These skills can be applied to market analysis, user behavior prediction, and data-driven decision-making.

8. AI Tools Reflection

(1) Usage

Use AI to assist in data cleaning, modeling code writing, and result interpretation.

(2) Performance evaluation

AI provided a clear code framework and explanations, which helped me get started with Python quickly.

(3) Learning takeaways

I learned how to transform technical results into business insights.

(4) Future and Ethics

The appropriate use of AI in academia and profession as an aid rather than a replacement, ensuring that results are transparent and explainable.

9. Appendix

(1) Data set

Name: Telco Customer Churn (IBM Sample Data)

Source: <https://www.kaggle.com/datasets/blatchar/telco-customer-churn>

(2) Code

① Data cleaning and EDA (cleaning and EDA.py):

Step 1: Import

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Step 2: Loading data

```
df = pd.read_csv("WA_Fn-UseC_-Telco-Customer-Churn.csv")
```

Step 3: View basic data information

```
print("First 5 rows of the dataset:")
print(df.head())
```

```
print("\nDataset info:")
print(df.info())
```

```
print("\nMissing values before conversion:")
print(df.isnull().sum())
```

Step 4: Data cleaning

Convert TotalCharges to numerical values

```

df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
# Check for missing values
print("\nMissing values after conversion:")
print(df.isnull().sum())
# Delete rows with missing values in TotalCharges
df = df.dropna(subset=['TotalCharges'])

# Step 5: Basic Statistics
print("\nChurn count:")
print(df['Churn'].value_counts())
print("\nChurn proportion:")
print(df['Churn'].value_counts(normalize=True))

# Step 6: Visualize churn rate
plt.figure(figsize=(6,4))
sns.countplot(x='Churn', hue='Churn', data=df, palette='Set2', legend=False)
plt.title("Customer Churn Count")
plt.show()

# Step 7: Analysis of the relationship between contract type and turnover
plt.figure(figsize=(8,5))
sns.countplot(x='Contract', hue='Churn', data=df, palette='Set1')
plt.title("Contract Type vs Churn")
plt.show()

# Step 8: Analyzing the relationship between monthly fees and churn
plt.figure(figsize=(8,5))
sns.histplot(df[df['Churn']=='Yes']['MonthlyCharges'], color='red', label='Churned Customers',
kde=True)
sns.histplot(df[df['Churn']=='No']['MonthlyCharges'], color='green', label='Retained Customers',
kde=True)
plt.legend()
plt.title("Monthly Charges vs Churn")
plt.show()

```

② Modeling and evaluation (model.py):

```

# Step 1: Import
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report, confusion_matrix, roc_auc_score

# Step 2: Loading and cleaning data

```

```

df = pd.read_csv("WA_Fn-UseC_-Telco-Customer-Churn.csv")
df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors='coerce')
df = df.dropna(subset=["TotalCharges"])
df = df.drop(['customerID'], axis=1)

# Step 3: Convert target variable to binary
df['Churn'] = df['Churn'].map({'Yes':1, 'No':0})

# Step 4: Converting categorical variables to numeric values      (One-Hot Encoding)
df_encoded = pd.get_dummies(df)

# Step 5: Divide into training and test sets
X = df_encoded.drop('Churn', axis=1)
y = df_encoded['Churn']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

# Step 6: Data standardization
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Step 7: Build a logistic regression model (increase the number of iterations)
model = LogisticRegression(max_iter=5000, solver='lbfgs')
model.fit(X_train_scaled, y_train)

# Step 8: Model prediction
y_pred = model.predict(X_test_scaled)

# Step 9: Model Evaluation
print("Classification Report:")
print(classification_report(y_test, y_pred))

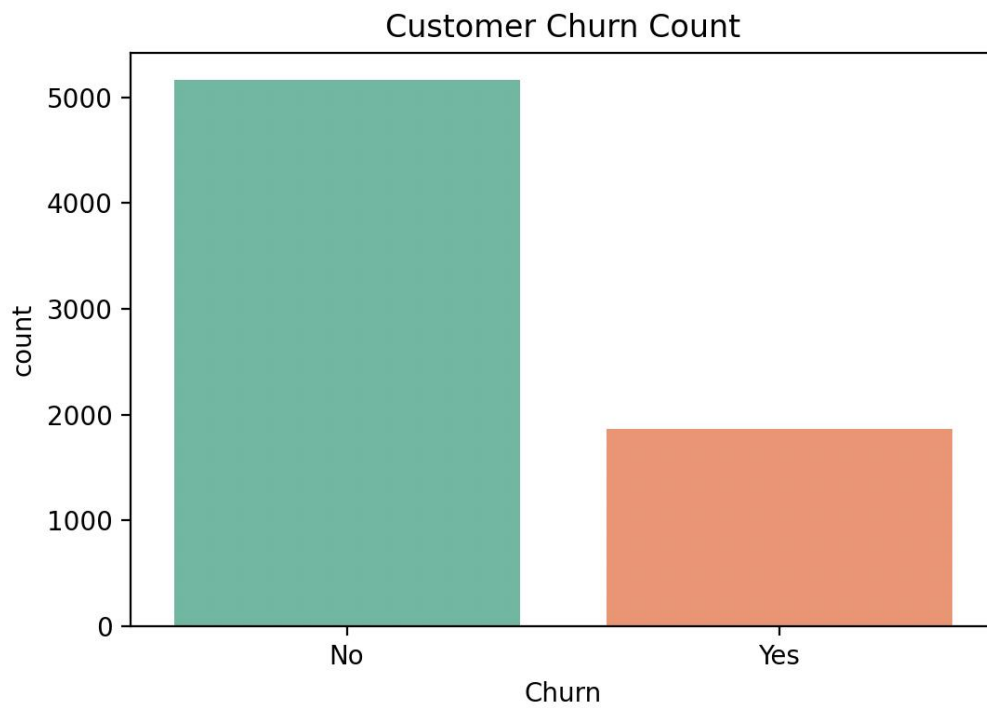
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

print("\nROC AUC Score:")
print(roc_auc_score(y_test, y_pred))

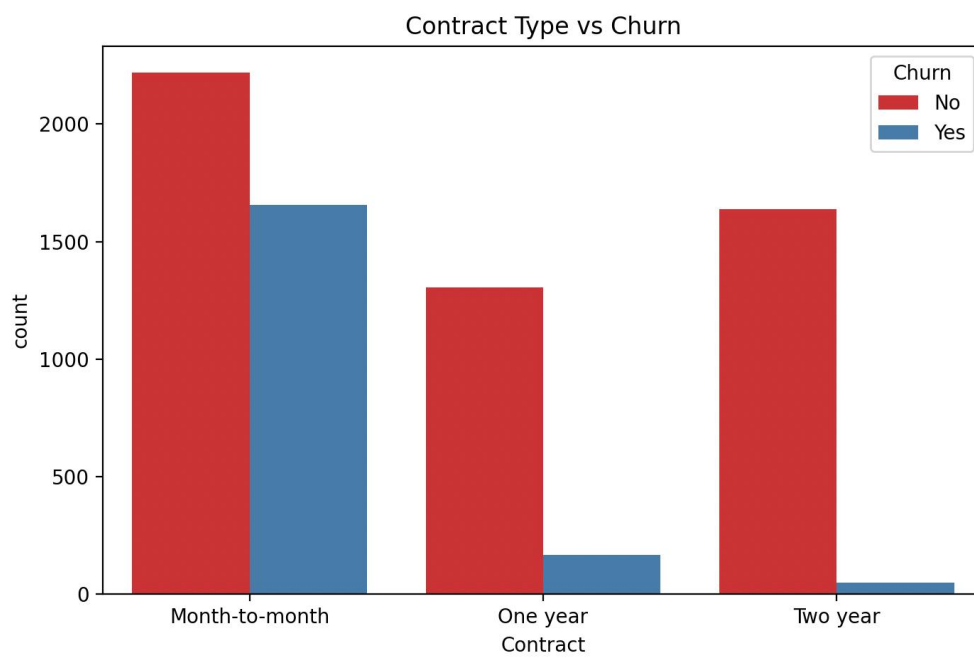
```

(3) Supplementary charts

- ① Customer Churn Count



② Contract type vs. churn



③ Monthly Charges vs Churn

